

Jongseo Lee

Vision and Learning Lab, Kyung Hee University, South Korea

Contact Information

E-mail: jong980812@khu.ac.kr

Personal Website: <https://jong980812.github.io>

Lab Website: <https://vll.khu.ac.kr/>

GitHub: <https://github.com/jong980812>

LinkedIn: www.linkedin.com/in/jongseo-lee-a551ab244

Research Interests

- Video understanding and action recognition
- Video-language multimodal learning (Video-LLMs)
- Video world models
- Explainable and interpretable AI (concept bottleneck models)
- Video class-incremental learning

Education

M.S. in Computer Science , Kyung Hee University, South Korea Advisor: Prof. Jinwoo Choi (Vision and Learning Lab) GPA: 4.43 / 4.5	Sep. 2023 – Aug. 2025
B.S. in Biomedical Engineering , Kyung Hee University, South Korea GPA: 4.23 / 4.5 (overall, including dual degree)	Mar. 2017 – Feb. 2023
B.S. in Electronics Engineering (dual degree), Kyung Hee University, South Korea	Mar. 2020 – Jul. 2023

Research Experience

Post-Master Researcher , Vision and Learning Lab, Kyung Hee University, South Korea Leading research on video-LLM evaluation/diagnosis and explainable video understanding; international collaborations with UC San Diego (Prof. Tsui-Wei Weng, TRACE) and Princeton University (Jihoon Chung, DeltaDirect).	Sep. 2025 – Present
Graduate Research Assistant , Vision and Learning Lab, Kyung Hee University, South Korea Researched video class-incremental learning and explainable action recognition; collaborated with Intel Labs (Kyle Min) and ETRI through funded research projects.	Sep. 2023 – Aug. 2025
Undergraduate Research Intern , Vision and Learning Lab, Kyung Hee University, South Korea Conducted research on video action recognition; co-led CAST (NeurIPS 2023) as a co-first author.	Feb. 2022 – Jul. 2023

Publications

(* equal contribution, † corresponding author)

International Conferences & Journals

1. **Which Way Did It Move? Diagnosing and Overcoming Directional Motion Blindness in Video-LLMs**
Jongseo Lee*, Hyuntak Lee*, Sunghun Kim, Sooa Kim, Jihoon Chung, Jinwoo Choi†
NeurIPS 2026 arXiv:2605.22823 / project / code
Also presented at the 2nd Workshop on Compositional Learning: Safety, Interpretability, and Agents (CompLearn), ICML 2026 (non-archival poster)
2. **TRACE: Trajectory-Aware Concepts for Explainable Video Understanding**
Wooil Lee, Junpyo Hong, Jongseo Lee, Tuomas Oikarinen, Tsui-Wei Weng, Jinwoo Choi†
NeurIPS 2026
3. **Disentangled Concepts Speak Louder Than Words: Explainable Video Action Recognition**
Jongseo Lee, Wooil Lee, Gyeong-Moon Park†, Seong Tae Kim†, Jinwoo Choi†
NeurIPS 2025, Spotlight (top 3.5%) arXiv:2511.03725 / code
4. **ESSENTIAL: Episodic and Semantic Memory Integration for Video Class-Incremental Learning**
Jongseo Lee*, Kyungho Bae*, Kyle Min, Gyeong-Moon Park†, Jinwoo Choi†
ICCV 2025, Highlight Presentation (top 2.9%) arXiv:2508.10896 / code
5. **CA²ST: Cross-Attention in Audio, Space, and Time for Holistic Video Recognition**
Jongseo Lee*, Joohyun Chang*, Dongho Lee, Jinwoo Choi†
IEEE TPAMI, 2025 arXiv:2503.23447

6. **PCBEAR: Pose Concept Bottleneck for Explainable Action Recognition**
Jongseo Lee, Wooil Lee, Gyeong-Moon Park[†], Seong Tae Kim[†], Jinwoo Choi[†]
[CVPR Workshop on XAI4CV 2025, Spotlight \(16.7%\)](#) arXiv:2504.13140
7. **CAST: Cross-Attention in Space and Time for Video Action Recognition**
Dongho Lee*, Jongseo Lee*, Jinwoo Choi[†]
[NeurIPS 2023](#) poster / code
8. **Metaverse Interface with Haptic and Rigid Sense Feedback at a Low Cost**
Jongseo Lee, Su Hyeon Kim, Sun Woong Jang, Jun Yeong Moon, Doug Young Suh[†]
[Journal of Appropriate Technology](#), 8(2), 2022 DOI:10.37675/jat.2022.00171

Preprints

9. **PCEvE: Part Contribution Evaluation Based Model Explanation for Human Figure Drawing Assessment and Beyond**
Jongseo Lee*, Geo Ahn*, Seong Tae Kim[†], Jinwoo Choi[†]
arXiv preprint, 2024 arXiv:2409.18260

Non-Archival Workshop Presentations

- **BiAdA: Holistic Video Recognition via Bidirectional Adapter Fusion**
ICCV 2025 Workshop on Long Video Foundations (LongVidFoundations), non-archival poster

Domestic Conferences (KIISE, 2024)

- Video Concept Bottleneck Model
- Efficient Video Class Incremental Learning via Class Token Re-Learning
- Audio-Video Cross Attention for Effective Video Action Recognition

Research Projects

Motivational Context Analysis of Interaction-Based Reactions in Video Funded research project with ETRI	2025
Classification and Explanation Models for ASD Screening Based on Drawing Video Understanding Funded research project with ETRI — related publication: PCEvE (arXiv:2409.18260)	2024
Deep Learning Models and Explainable AI for ASD Screening Drawing Data Funded research project with ETRI	2023
Rule-Based Group Activity Recognition Video Dataset (AI Hub) National dataset construction project in collaboration with GAZZILABS — developed and validated AI recognition algorithms	2022

Honors & Awards

- | | |
|--|------|
| • Spotlight Presentation, NeurIPS 2025 (top 3.5% of accepted papers) | 2025 |
| • Highlight Presentation, ICCV 2025 (top 2.9% of accepted papers) | 2025 |
| • Spotlight, CVPR Workshop on Explainable AI for Computer Vision | 2025 |

Technical Skills

Programming: Python, C/C++ **Frameworks:** PyTorch
Systems: Large-scale video/multimodal model training and evaluation

Academic Service

Reviewer:

- NeurIPS 2026
- AAAI 2025
- CVPR Workshop on Explainable AI for Computer Vision (XAI4CV) 2026
- 2nd Workshop on Compositional Learning: Safety, Interpretability, and Agents (CompLearn), ICML 2026

References

Prof. Jinwoo Choi — Kyung Hee University, South Korea (M.S. advisor)
Prof. Gyeong-Moon Park — Korea University, South Korea (co-author of DANCE, ESSENTIAL, PCBEAR)
Prof. Seong Tae Kim — Kyung Hee University, South Korea (co-author of DANCE, PCBEAR, PCEvE)
Prof. Tsui-Wei Weng — UC San Diego, USA (co-author of TRACE)
Kyle Min — Oracle, USA (co-author of ESSENTIAL; formerly Intel Labs)

Contact information available upon request.